Daniel Hardesty Lewis

Forward Deployed Engineer – full-stack, enterprise operations, AI & automation

Engineer and repeat founder specializing in bridging product engineering and enterprise customers. Built full-stack platforms (Python, React, SQL), automated operational workflows, and shipped customer-facing tooling from scoping through launch.

Core Expertise

8 years	**Full-Stack Engineering**: Build and ship production web applications end-to-end using Python, React, SQL, and Postgres; own from data pipeline through user-facing delivery
4 years	**Client Engineering & Onboarding**: Scope requirements, triage requests, and manage technical integrations with enterprise stakeholders across government, financial, and engineering verticals
8 years	**Process Automation**: Identify manual bottlenecks and replace them with automated workflows, AI tooling, and repeatable playbooks to scale operations
8 years	**Communication & Education**: Translate technical concepts for non-technical audiences; teach, document, and build institutional knowledge across teams and clients
8 years	**Technical Leadership**: Drive cross-functional execution across engineering, product, and external partners; mentor engineers through design reviews, code reviews, and instruction

Technologies

Core stack	Python; SQL; TypeScript; React; Postgres; Redis; Elasticsearch; Supabase; FastAPI
AI & automation	LangChain; LangGraph; DSPy; RAG; FAISS; PyTorch; scikit-learn; Airflow
Data	Pandas; Polars; PyArrow; Parquet; BigQuery; PostGIS; ETL pipelines
Infrastructure	Linux; Git; Docker; Kubernetes; Terraform; CI/CD; monitoring & observability
Additional	CUDA; C++; Ray; Dask; Spark

Experience

Professional

Apr. 2025 – Present
Co-founder, *Homecastr*, New York, NY
Consumer real-estate pricing platform; owned end-to-end workflow from source data to production web delivery.
- Built and shipped a **full-stack web application** from zero to production using React, Python/FastAPI, and Postgres.
- Deployed ML pricing models on tabular property records; designed evaluation frameworks and **automated data pipelines** for continuous model updates.
- **Scoped product requirements**, ran user interviews, and iterated on positioning through feedback from a16z, MetaProp, and other enterprise stakeholders.

Nov. 2024 – Present
Co-founder, *PoliBOM*, Seattle, WA
AI tariff mitigation product for manufacturers; led **customer-facing scoping, demos, and technical onboarding** for enterprise prospects.
- Built an **AI-powered automation workflow** to replace manual BOM analysis, reducing tariff compliance triage from days to minutes.
- Directed a **conversational AI interface** enabling non-technical decision-makers to retrieve BOMs and receive proactive compliance alerts without engineering support.
- Ran **customer discovery and live demos** with enterprise stakeholders including KPMG; **scoped integration requirements** and documented onboarding checklists.
- Executing ongoing discovery with importers, exporters, and tariff economists to iterate on workflows, pricing, and product-market fit.

Nov. 2023 – Present
Founder, *Summit Geospatial*, New York, NY
Geospatial data platform serving engineering and government clients.
- Built the **web distribution platform end-to-end**: data pipeline, tiling, hosting, and delivery UX.
- Managed commercial licensing discussions with research organizations and AI companies.

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dhardestylewis • *Google Scholar*

Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- **Triaged and scoped enterprise requests** across 6+ state and federal agencies (NOAA, NSF, USACE, GLO), translating technical requirements into RFP and MOU frameworks with clear launch checklists.
- Served as primary **technical point of contact** for the $40M Texas Disaster Information System (TDIS), coordinating between engineering, agency stakeholders, and end-users.
- Built **repeatable processes and documentation** for data onboarding, enabling partner agencies to self-serve integrations with the platform.
- Managed multi-million-dollar compute budgets and orchestrated distributed workloads, identifying opportunities to **automate** pipeline steps and reduce manual intervention.
- **Managed relationships** with federal, state agencies, and universities including NOAA, NSF, USACE, Texas GLO, USC, UT, and A&M.
- **Evaluated and accepted** $10M+ in computing resource requests as Technical Reviewer for Frontera Doctoral Fellows and Large-Scale Computing Pathways, triaging proposals against capacity and impact criteria.
- Co-founded summitgeospatial.com to commercialize platform outputs, owning the customer-facing product from data pipeline through web delivery.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in year-long DARPA World Modelers program, coordinating across distributed engineering and research teams to deliver integrated modeling workflows.
- Received recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

Research

Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirsa**, *Columbia University*, New York, NY
Developed production ML models for financial data; built evaluation and explainability frameworks for industry-sponsored research.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA
Integrated hydrological models into DARPA's scientific orchestration platform.

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

- Designed and taught graduate-level courses including *Machine Learning for the Geosciences* for Petrobras geoscientists and petroleum engineers, *Intelligent Systems*, and *Scientific Computation*. Developed course materials, documentation, and hands-on exercises.

Select Publications

2 peer-reviewed publications in ACM (TiiS 2021, IUI 2019); full list on Google Scholar.

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

Languages

Spanish Native Proficiency

French Professional Proficiency

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